Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution — Reading • 4 min
- Gaussian Distribution: Foundations and Applications — Reading • 15 min
- **Multivariate Gaussian Distribution** — Reading • 10 min
- Conditional and Marginal Multivariate Gaussian Distributions — Reading • 45 min
- Product of Gaussian Densities — Reading • 26 min

Lesson 3: Inference Techniques

- Bayesian Inference — Reading • 1h 10m
- Latent-Variable Models — Reading • 50 min
- Check Your Knowledge: Inference Techniques — Graded Assignment • 10 min

Module 4
Introduction to Model Evaluation and Optimization

Multivariate Gaussian Distribution

0:33 / 1:17 2x

The multivariate Gaussian distribution generalizes the univariate normal distribution to multiple variables and describes the joint distribution of a vector of random variables.

$$f(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^k |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

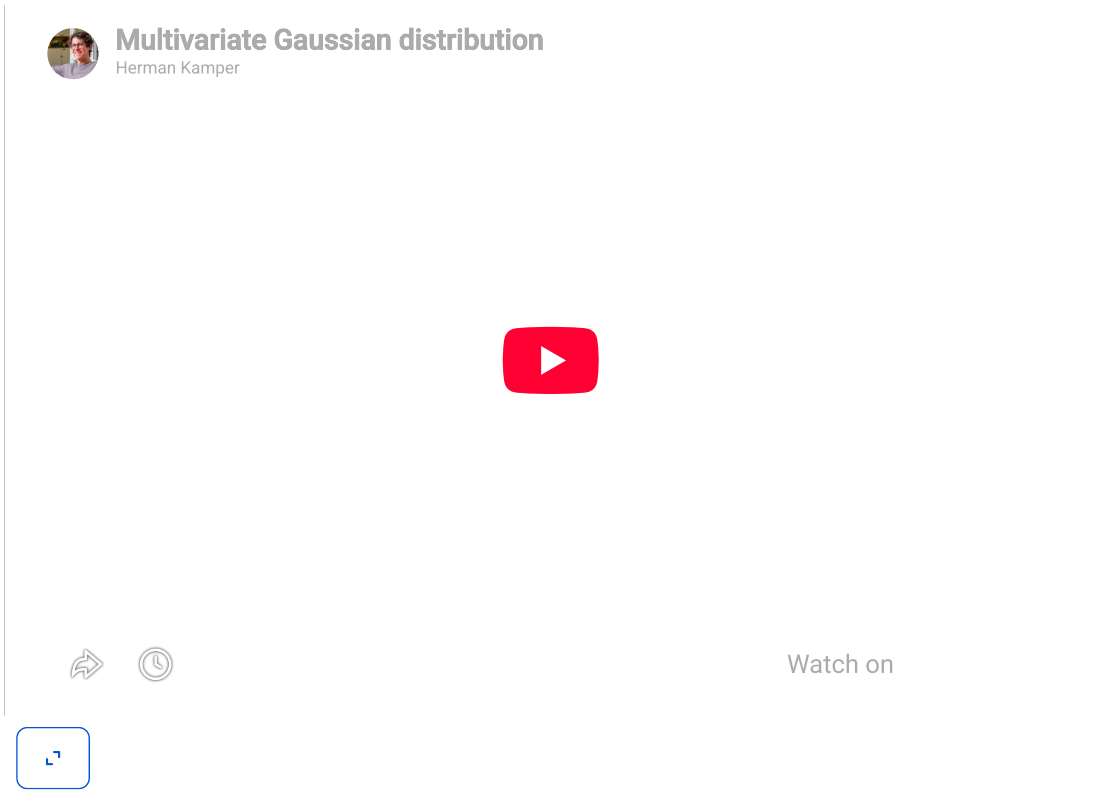
Where:

- x is a k-dimensional random vector.
- μ is the mean vector of dimension k.
- Σ is the covariance matrix of dimension $k \times k$.

Characteristics:

- The mean vector μ indicates the center of the distribution.
- The covariance matrix Σ describes the shape and orientation of the distribution.
- If the covariance matrix is diagonal, the variables are independent.

This video explains the mean vector and covariance matrix in a multivariate Gaussian distribution. Using examples, you'll learn how the mean vector locates the distribution's center and how the covariance matrix affects its spread, orientation, and correlations between variables. The video illustrates these concepts with two-dimensional distributions, showing how different covariance structures change the shape and alignment of the distribution.



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